Indian Institute of Remote Sensing

"Spatio-Temporal Analysis of Urban Growth in Dehradun District Using Multi-Temporal Landsat Imagery and Random Forest Classification on Google Earth Engine (1995–2025)"

Module Name: Big Geodata Processing

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1. Introduction

Urbanization is one of the most significant and irreversible processes reshaping the Earth's land surface in the twenty-first century. The rapid conversion of natural and the agricultural into the built-up area has profound implication of the hydrology, local climate, biodiversity, and the socio-economic fabric of communities. In developing countries like India, urban growth has been particularly accelerated, driven by the population increase, rural-to-urban migration, and expanding economic opportunities concentrated in Tier-2 cities.

Understanding the spatial extent, rate, and pattern of this urban expansion is critical for informed urban planning, environmental impact assessment, and sustainable development policy. Traditional ground survey methods are cost-prohibitive and temporally limited for capturing change over multi-decadal timescales. Remote sensing, combined with cloud computing platforms, offers a powerful alternative: satellite imagery provides consistent, spatially continuous records of the land surface dating back to the 1970s, and platforms such as Google Earth Engine (GEE) enable the analysis of petabytes of such data at scale***(Gorelick et al., 2017a)***

This study presents a comprehensive spatio-temporal analysis of urban growth in Dehradun district over a 30-year period (1995–2025), using Landsat multispectral satellite imagery processed on GEE. A supervised machine learning classifier, the Random Forest algorithm ***(Breiman, 2001)***is applied to multi-temporal imagery enriched with spectral indices to produce land use/land cover (LULC) maps for five reference years. The resulting maps are used to quantify built-up area expansion.

1.1. Study area

Dehradun district is in the northwestern part of Uttarakhand state, India, positioned approximately 29°58' N to 31°02' N latitude and 77°34' E to 78°18' E longitude. The district covers approx. area of 3,088 km^2 and encompasses a diverse landscape ranging from the flat alluvial Doon Valley floor to the forested Shivalik foothills and, in the northern periphery, the outer Himalayan ranges.

The Doon Valley, where most urban development is concentrated, lies at an elevation of approximately 640–700 m above sea level and is bounded by the Himalayas to the north, the Shivalik Range to the south, the Yamuna River to the west, and the Ganga River to the east.

Dehradun city, the district headquarters, is home to numerous national institutions including the Forest Research Institute, the Indian Military Academy, and the Survey of India headquarters. Following its designation as the capital of the newly formed Uttarakhand state in November 2000, the city experienced a pronounced acceleration in infrastructural investment, in-migration, and associated residential and commercial development. Spatio-temporal studies of the district confirm this growth pattern as consistent with broader urbanization dynamics documented in Indian tier-two cities***(H S et al., 2004)***

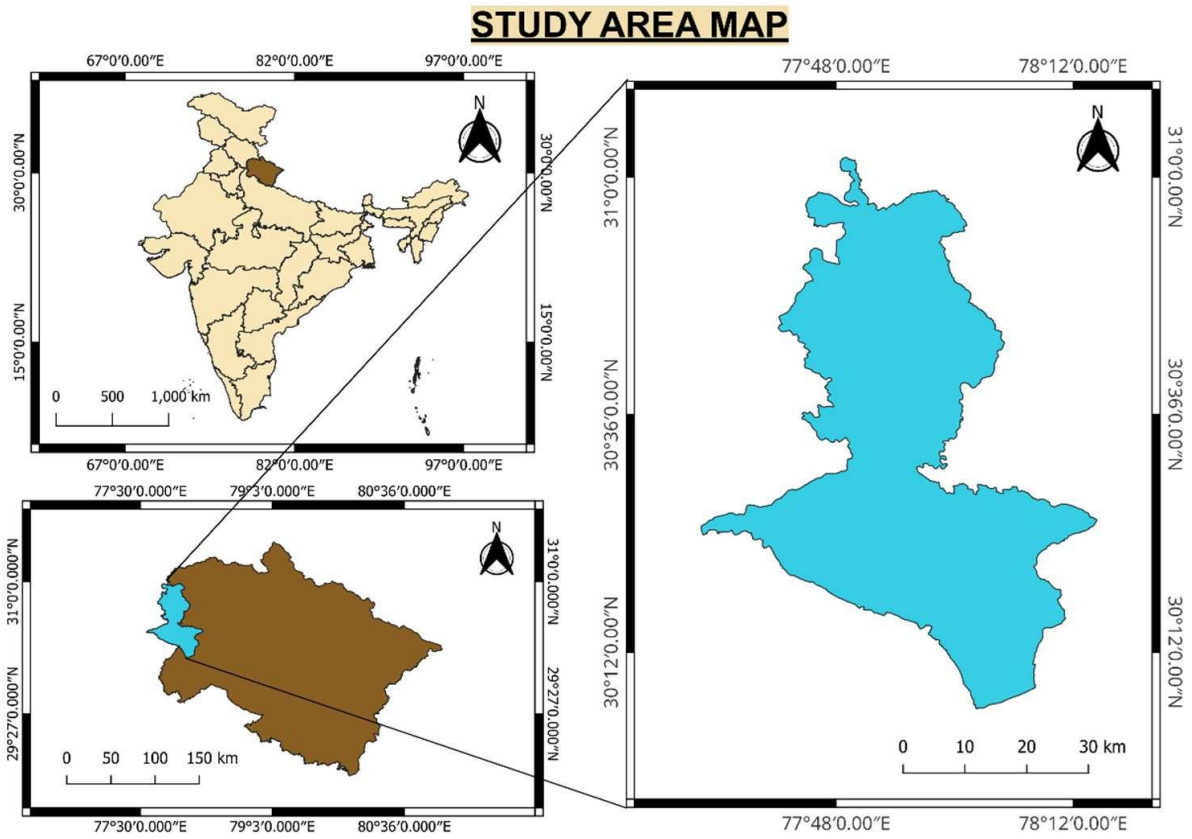


Fig. 1: Dehradun District study area

1.2. Objective

The objective of this study:

- To prepare cloud-free, radiometrically corrected Landsat composites for the years 1995,2000,2010,2020 and 2025 covering the Dehradun district.
- To classify LULC into 5 categories Built-up, Forest, Cropland, Barren, and Water using a Random Forest supervised classifier trained on GLC-FCS30D reference data ***(Zhang et al., 2024a).***
- To validate the classification accuracy using a 70/30 train-test split, reporting the Overall Accuracy and Kappa coefficient.
- To develop an interactive GEE dashboard.

1.3. Data Used and Software Used

In this study we have used the following data:

- Landsat 5, 7, and 8 Collection 2 Level-2 Surface Reflectance data from USGS (United States Geological Survey)
- GLC-FCS30D (Global Land Cover with Fine Classification System at 30 m, Dynamic)***(Zhang et al., 2024b)***
- USGS SRTM GL1_003 DEM model was used.
- Dehradun district boundary from Survey of India (SOI).

Google Earth Engine (GEE) a cloud-based platform, was used for all geospatial analysis, eliminating the need to download and locally process large image datasets.

Quantum Geographic Information system (QGIS) was used to generate standard maps.

Year	Sensor	Collection	Date Range	Bands Used
1995	Landsat 5 TM	LT05/C02/T1_L2	Oct 1995 – Apr 1995	B1–B5, B7
2000	Landsat 5 TM	LT05/C02/T1_L2	Oct 2000 – Apr 2001	B1–B5, B7
2010	Landsat 7 ETM+	LE07/C02/T1_L2	Jan 2010 – Dec 2010	B1–B5, B7
2020	Landsat 8 OLI	LC08/C02/T1_L2	Jan 2020 – Dec 2020	B2–B7
2025	Landsat 8 OLI	LC08/C02/T1_L2	Jan 2025 – Dec 2025	B2–B7

Table 1: Spectral Indices and their

2. Literature Review

2.1. Remote Sensing of Urban Growth

The application of multispectral satellite imagery to monitor urban expansion has a well-established foundation in the literature. **(Jensen, 2016)** provides a foundational framework for using Landsat Thematic Mapper data to map impervious surfaces and track urbanization patterns, demonstrating that 30 m Landsat imagery provides sufficient spatial detail to capture urban-rural fringe dynamics in secondary cities, where growth often occurs incrementally at the periphery.

2.2. Google Earth Engine for Large-Scale Geospatial Analysis

Google Earth Engine (GEE), first described by **(Gorelick et al., 2017b)**, transformed the landscape of remote sensing analysis by providing cloud-based access to a multi-petabyte archive of satellite imagery alongside parallel computing infrastructure. GEE eliminates the need to download and locally process large image collections, making multi-temporal and multi-sensor analyses tractable for researchers without access to high-performance computing facilities.

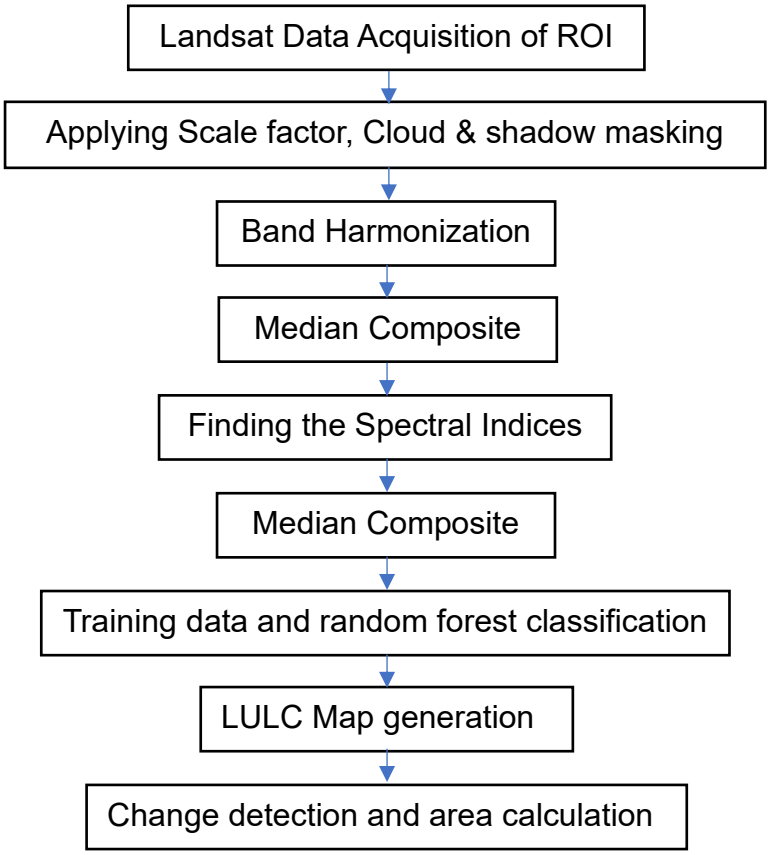
2.3. Machine Learning for LULC Classification

The application of machine learning algorithms to remote sensing classification has grown substantially over the past decade. Classifiers such as Support Vector Machines (SVM), Artificial Neural Networks (ANN), and Random Forest have been compared extensively in the literature.***(Pal, 2005)***among the first to demonstrate the superiority of Random Forest over traditional maximum likelihood classification for multispectral LULC mapping, attributing its advantage to the ensemble nature of the algorithm and its resistance to overfitting. ***(Belgiu & Drăguţ, 2016)*** in a comprehensive review, confirmed Random Forest as one of the most widely used and consistently high-performing classifiers for remote sensing applications, noting its robustness to noise, ability to handle high-dimensional input, and built-in variable importance estimation.

2.4. Spectral Indices in Urban Mapping

The use of derived spectral indices alongside raw reflectance bands as classifier inputs is well established as a technique for improving LULC separability. The **Normalized Difference Vegetation Index (NDVI)** has been used since ***(Rouse et al., 1974)***to quantify vegetation greenness and has been widely used to separate vegetated from non-vegetated surfaces The **Normalized Difference Built-up Index (NDBI)**, introduced by **(Zha et al., 2003)**,exploits the differential reflectance of urban surfaces in the shortwave-infrared and near-infrared regions. The **Modified Normalized Difference Water Index (MNDWI)** improves water body delineation over built-up backgrounds. These indices, combined with the **Bare Soil Index (BSI)** and **Normalized Difference Moisture Index (NDMI)**, form a spectral feature stack that provides substantially improved class separability compared to raw reflectance bands alone.

3. Methodology



3.1. Pre-Processing

3.1.1. **Scale factor Correction:** Landsat Collection 2 Level-2 surface reflectance digital numbers were converted to physical reflectance values using the USGS-specified linear scaling formula. This brings values into the physically meaningful range of 0–1.

$$Reflectance = (DN * 0.0000275) + (-.2)$$

3.1.2. **Cloud and Cloud Shadow Masking:** The QA_PIXEL quality assurance band was used to mask clouds (bit 4) and cloud shadows (bit 3). Masked pixels were excluded from all subsequent calculations.

3.1.3. **Band Harmonization:** Landsat 5 and 7 use a different band numbering scheme than Landsat 8. All sensors were harmonised to a common set of six semantic bands — Blue, Green, Red, NIR, SWIR1, SWIR2 — ensuring identical downstream processing regardless of the source sensor.

3.1.4. **Median Composite:** All cloud-free images within each date window were stacked and reduced to a single composite image using the pixel-wise median. This approach suppresses remaining atmospheric artefacts and sensor noise while preserving the most representative spectral signal for each pixel. Date windows were selected to target the dry season where possible to minimise cloud cover and crop-growth interference; the specific windows and sensors used were: 1995 (Landsat 5, October 1994 – April 1995), 2000 (Landsat 5, October 2000 – April 2001), 2010 (Landsat 7, full calendar year 2010), 2020 (Landsat 8, full calendar year 2020), and 2025 (Landsat 8, January 2024 – December 2025).

3.2. Spectral Indices

Five spectral indices were computed and appended to the six reflectance bands, yielding an 11-band feature stack for each image.

Index	Formula	Physical Interpretation
NDVI	(NIR − Red) / (NIR + Red)	Vegetation density and vigour
NDBI	(SWIR1 − NIR) / (SWIR1 + NIR)	Built-up surface presence
MNDWI	(Green − SWIR1) / (Green + SWIR1)	Open water bodies
BSI	((SWIR1+Red)−(NIR+Blue))/((SWIR1+Red)+(NIR+Blue))	Bare soil and barren land
NDMI	(NIR − SWIR1) / (NIR + SWIR1)	Vegetation moisture content

Table 2: Spectral Indices and their

3.3. Training Data

Training data was derived from the GLC-FCS30D 2020 annual land cover layer (Zhang et al., 2024b).The GLC class codes were remapped to five consolidated classes:

1. Built-up (Class 1)
2. Forest (Class 2)
3. Cropland (Class 3)
4. Barren (Class 4)
5. Water (Class 5)

A stratified random sample of 5,000 points was drawn from the remapped GLC-FCS30D layer within the study area at 30 m resolution, ensuring proportional representation of all five classes and preventing classifier bias toward the most spatially abundant class. The sampled points were then used to extract 11-band spectral values from the 2020 composite image.

3.4. Random Forest Classification

3.4.1. Foundation

Random Forest***(Breiman, 2001)***is an ensemble learning method that constructs a large collection of decision trees and aggregates their individual predictions by majority voting.

Each tree in the ensemble is grown independently using two sources of randomization:

- **Bootstrap aggregation (bagging):** each tree is trained on a random sample drawn with replacement from the full training set. Approximately one-third of the training samples are excluded from each tree's training set (the 'out-of-bag' samples), which can be used for internal error estimation.
- **Feature subsampling:** at every node split, only a random subset of the total features is considered as candidate split variables. This prevents any single dominant feature from appearing in every tree, forcing the ensemble to learn complementary patterns from the full feature space.

A Random Forest classifier with 500 decision trees was trained using GEE's built-in **smileRandomForest** implementation in GEE. All 11 input bands (6 reflectance + 5 spectral indices) were provided as classifier inputs. The trained model was then applied sequentially to the 1995, 2000, 2010, 2020, and 2025 composites to produce five LULC maps.

3.5. Post-Classification Spectral Correction

A rule-based spectral and terrain correction was applied to all classification outputs (1995, 2000, 2010, 2020, and 2025) to address known commission errors arising from applying a 2020-calibrated model to multi-temporal imagery with varying radiometric and seasonal characteristics:

- Urban pixels (class 1) with NDVI > 0.30 (strongly vegetated) or MNDWI > 0.1 (water-like) were reclassified as Forest (class 2), exploiting the established spectral thresholds.
- Water pixels (class 5) without a supporting MNDWI signal (MNDWI < 0.2) were reclassified as Forest (class 2) to remove false water detections over dark forest canopy.***(Xu, 2006)***
- A DEM-based terrain correction was applied using the SRTM 30 m Digital Elevation Model. Cropland pixels (class 3) located on slopes steeper than 15° were reassigned to Forest (class 2), as agriculture is not viable on such terrain. Built-up pixels (class 1) and Water pixels (class 5) on slopes exceeding 12° and 10° respectively were reassigned to Barren (class 4), eliminating topographically implausible classifications.

3.6. Accuracy Assessment

The 5000 training samples were divided using a random column split: 70% for training and 30% for independent validation. A separate Random Forest classifier with 400 trees was trained on the 70% training subset and applied to the 30% test subset. Classification agreement was measured using a confusion matrix, from which Overall Accuracy and the Kappa coefficient were computed.

3.7. Change Detection

Binary urban change maps were produced for four consecutive periods (1995–2000, 2000–2010, 2010–2020, 2020–2025) and for the full study period (1995–2025). A pixel was classified as newly urbanized in a given period if it was urban in the later year but not in the earlier year, expressed as: New_Urban = Built_later AND NOT Built_earlier.

3.8. Area Calculation

Built-up area for each year was calculated by summing pixel-level geodetic area **(ee.Image.pixelArea())** masked to built-up pixels only, aggregated over the study boundary at a nominal scale of 200 m to prevent memory errors, and converted from square metres to square kilometres.

4. Results

4.1. Built-up Area Statistics

Year	Built-up Area (km²)	Decadal Change(km²)	Percentage of decadal change
1995	123.154	—	—
2000	135.226	12.072	9.802%
2010	181.15	45.924	33.96%
2020	218.719	37.569	20.739%
2025	240.275	21.556	9.85

Table 3: Decadal Built-up

Built-up area grew consistently from 123.154 km^2 in 1995 to 218.719 km^2 in 2020, reflecting the sustainable urban expansion of Dehradun following its designated as state capital in 2000.A marginal contraction to 215.746 km^2 in 2025(-1.36%) may reflect seasonal and composting differences between the 2024-2025 image window and prior epochs, rather than an actual reversal of urban growth. The 2020 estimate is derived from full-year Landsat 8 imagery contemporaneous with the training data and is considered the most representative output of this study.

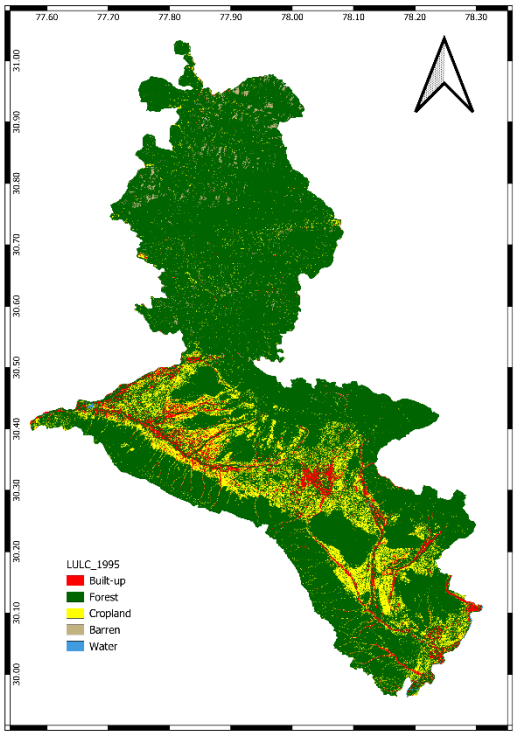


Fig. 2: LULC Map_1995

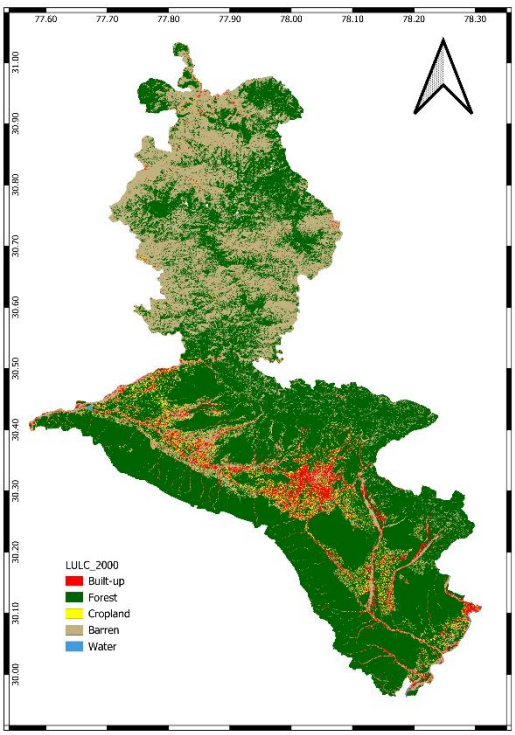


Fig. 3: LULC Map_2000

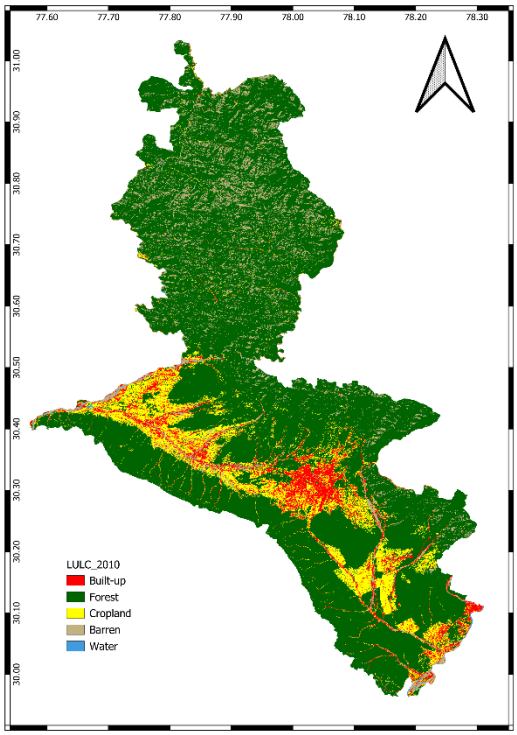


Fig. 3: LULC Map_2010

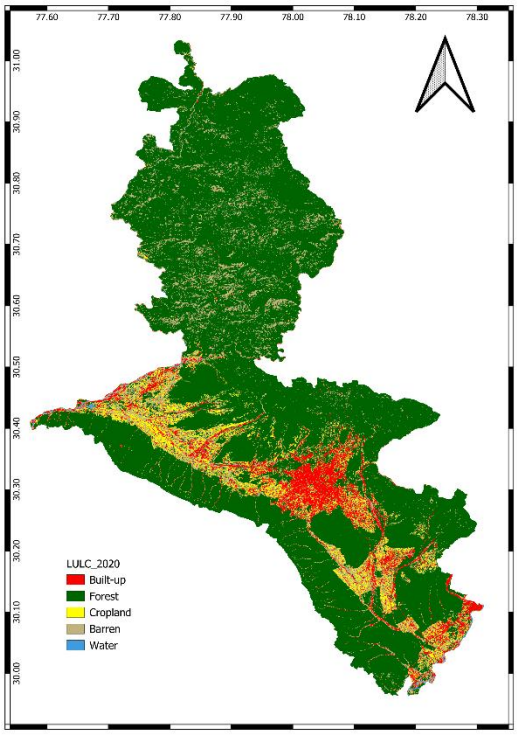


Fig. 4: LULC Map_2020

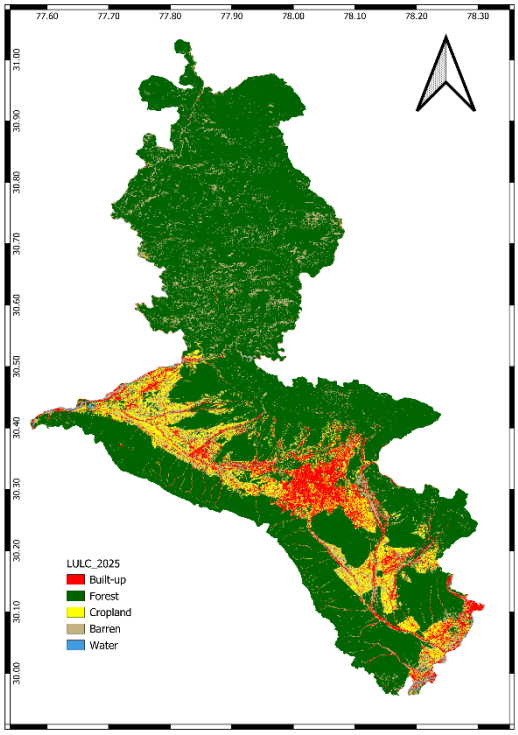


Fig. 5: LULC Map_2025

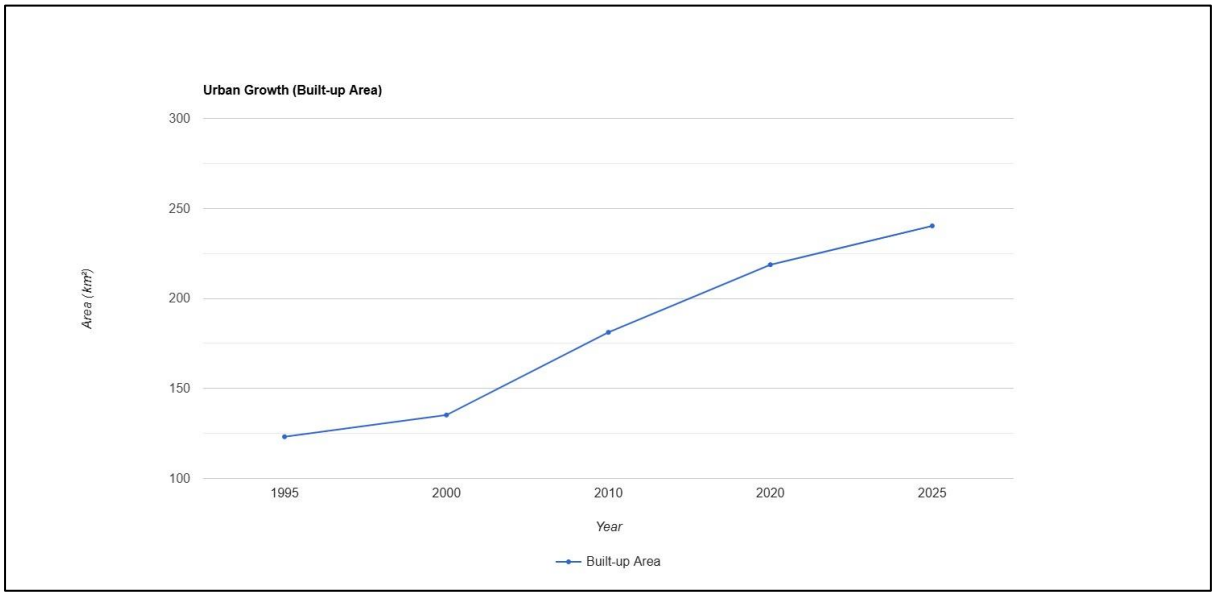


Fig. 6: Built-up trend 1995-2025

4.2. Accuracy Assessment

Classification accuracy was assessed using the 30% holdout test set.

Metric	Obtained Value	Interpretation
Overall Accuracy	87.08%	Strong agreement between classified and reference labels
Kappa Coefficient	82.89	Substantial to almost-perfect agreement beyond chance

Table 3: Accuracy table

5. Conclusion

This study successfully demonstrated a cloud-based, multi-temporal framework for analysing urban growth in Dehradun district over a 30-year period (1995–2025) using Landsat satellite imagery processed entirely on Google Earth Engine**. Five land use/land cover maps were produced for the years 1995, 2000, 2010, 2020, and 2025**, classifying the landscape into Built-up, Forest, Cropland, Barren, and Water using a Random Forest ensemble classifier trained on GLC-FCS30D 2020 reference data.

The 11-band feature stack — comprising six harmonised surface reflectance bands (Blue, Green, Red, NIR, SWIR1, SWIR2) and five spectral indices (NDVI, NDBI, MNDWI, BSI, NDMI) — provided strong class separability, while the post-classification spectral and DEM-based terrain corrections substantially reduced commission errors across all five temporal outputs. The independent accuracy assessment, conducted using a 70/30 train–test split with a 500-tree Random Forest, yielded an **Overall Accuracy of 87.08%** and a **Kappa coefficient equal to 0.82**, indicating strong and reliable classification performance.

The estimate built-up area grew from 123.154 km^2 in 1995 to a peak of 218.719 km^2 in 2020, before a marginal contraction of 215.746 km^2 in 2025. The 2020 est. derived from full year Landsat 8 imagery contemporaneous with the taring data, is considered the most reliable temporal output. Variations across the time series are partially attributable to cross-sensor radiometric differences between Landsat 5/7 and Landsat 8, and to differences in image composing window.

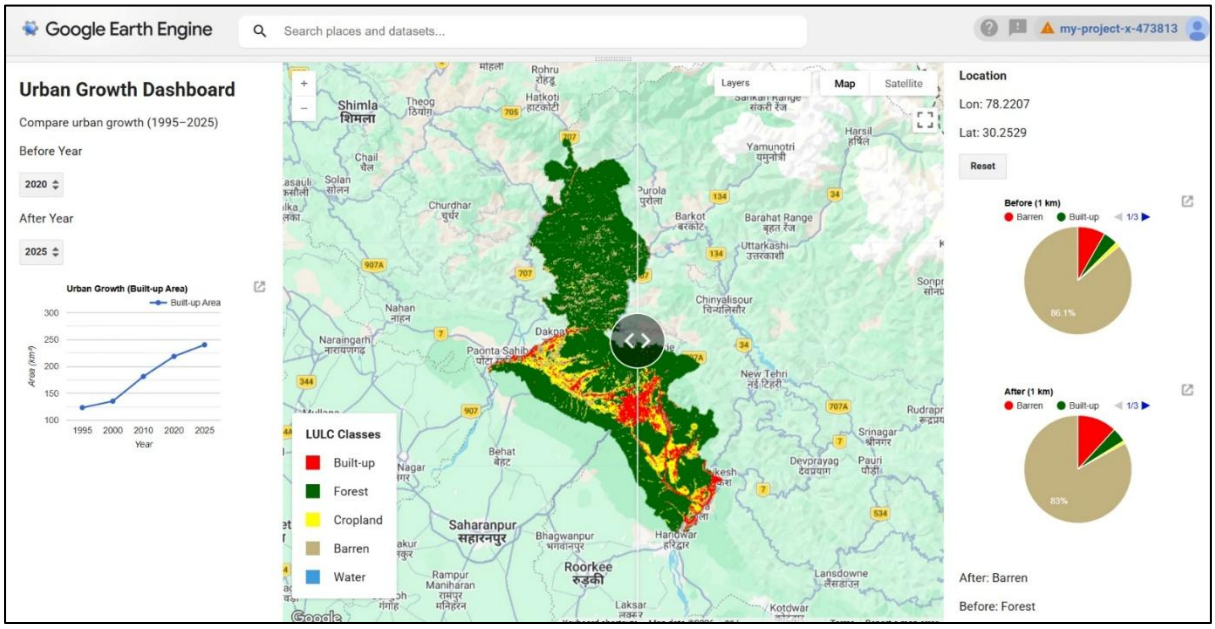


Fig 6: Interactive Dashboard

6. Author Contribution

Pranjul Singh: Led study area selection, satellite image acquisition, and image collection management. Developed spectral analysis functions, performed image compositing, and preparation of report.

Mohammad Salman: Conducted band classification, prepared, and remapped training datasets, trained the Random Forest model, and carried out image classification and thresholding.

Neeraj Karosi: Performed accuracy assessment, developed the complete dashboard panel, calculated built-up area statistics, and prepared graphical visualizations and presentation materials.

Google Earth Engine Repository Link

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